Ishaan Baru

Jammu, J&K | **+91 6006854030** | **ishaanbaru33@gmail.com |** GitHub | LinkedIn | Portfolio

Education

M.Tech in Computer Science | University of Jammu (Expected 2028) | MCA | MIET Jammu (2026) | B.Sc | IGNOU (2023)

Skills

Languages: C, C++, Python, SQL, Java, Bash, JavaScript

Systems & Architecture: Linux/POSIX, Win32 API, Multithreading, Thread Affinity & Synchronization, Hardware Telemetry, CPU Cache/TLB Behavior

Backend & Data Infrastructure: FastAPI, MySQL, PostgreSQL, Redis, ChromaDB, FAISS, Docker, REST APIs

Tools & Testing: Git, GCC/MinGW, Pytest, Pandas, NumPy, Linux CLI

Systems Research

"The Energy Fallacy of Thread Preemption: Empirical Quantification of Context Switching Overhead"

Author: Ishaan Baru | Department of Computer Science & IT, University of Jammu

- Evaluated thread preemption costs on an AMD Ryzen 5 5625U by pinning an identical 100M bitwise workload to Core 0 under two regimes: contiguous execution vs. 20,000 synchronized context switches.
- Telemetry (AMD SMU package power, Windows typeperf) demonstrated a 7.49× execution slowdown (14.75 s to 110.50 s) and a **635.9% surge in cumulative energy dissipation** (55.96 J to 411.86 J) despite a flat instantaneous power draw (3.79 W vs. 3.73 W).
- Proved that thread preemption overhead manifests primarily as cumulative energy waste rather than elevated instantaneous thermal dissipation.

Projects

Production-Grade Hybrid RAG Knowledge Engine | *Python, FastAPI, ChromaDB, Redis, Pytest, Docker*

- Built an end-to-end Retrieval-Augmented Generation (RAG) service to ingest, index, and query multi-format technical documentation with <180ms p95 retrieval latency.
- Engineered a hybrid search pipeline combining BM25 sparse keyword matching with dense vector embeddings, boosting top-5 retrieval recall by **28%** over pure semantic search.
- Implemented recursive semantic chunking and Redis caching for recurring embeddings, reducing redundant model inference calls and vector database compute by **42%**.
- Authored automated evaluation test suites using Pytest to systematically measure retrieval faithfulness, context precision, and guard against hallucinations.

Concurrent Transactional Ledger & Idempotent API | *Python, FastAPI, PostgreSQL, Docker, Pytest*

- Designed an ACID-compliant double-entry ledger backend processing concurrent inter-account transfers with zero balance drift.
- Enforced idempotency key verification and row-level locking to prevent duplicate debiting under network retries and concurrent race conditions.
- Structured Pydantic validation layers and custom domain exceptions to ensure non-negative balances and atomic rollbacks upon failure.
- Authored unit and integration test suites simulating payload tampering and depleted balances, maintaining **>90% branch coverage**.

Student Management System & Database Engine | *Python, MySQL, Tkinter*

- Engineered a desktop system managing student records and enrollments using parameterized MySQL queries and indexed keys to guarantee data integrity.

Experience

Backend Sales Operations Coordinator | Planet Spark|May *2024 – Dec 2024*

- Managed the order processing lifecycle across logistics, finance, and CRM systems, maintaining relational record integrity across large-scale datasets.
- Built automated reporting spreadsheets with advanced indexing formulas.

Certifications

IBM: Minor Degree / Certification in AI/ML **| Google:** Data Analytics Professional Certificate **| Infosys SpringBoard:** Java & SQL Developer Certifications